

Sandipto Roy

Last updated in August 2026

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Education

National Institute of Technology, Durgapur

Aug 2023 – Present

Bachelor of Technology in Computer Science and Engineering

- CGPA: 9.64/10

Jangipur High School

April 2021 – April 2023

Higher Secondary Education (WBCHSE)

- Percentage: 91.6%

Technical Skills

Languages & Backend: Java, Spring Boot, C++, Python, Node.js

Data & Messaging: PostgreSQL, MongoDB, Neo4j, Cassandra, Redis, Apache Kafka, RabbitMQ

Cloud, DevOps & Operations: Kubernetes, Docker, Terraform, GitLab, CI/CD, NGINX, Prometheus, Grafana, Loki

AI, APIs & Frontend: LangGraph, Mem0, Qdrant, GraphQL, gRPC, WebSockets, REST APIs, React, Next.js

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems, System Design

Experience

Summer Analyst – Goldman Sachs Services Private Limited

May 10 – July 05, 2026

- Architected a FastAPI orchestrator for an Agentic AI workflow, containerizing an ML model and establishing a 5-stage GitOps CI/CD pipeline on Kubernetes via ArgoCD for end-to-end orchestration.
- Engineered migration of 5 exception calculation engines from legacy systems to modern platforms using spec-driven development with GitHub Spec.
- Optimized a real-time React/Redux platform by resolving memory leaks, improving Interaction to Next Paint (INP) by 99.1% and LCP by 49%.

Research Intern – IIT Bombay (IASc-INSa-NASI Summer Research Fellowship)

15 May – 10 July, 2025

- Developed a **Federated Retrieval-Augmented (RAG)** system for **privacy-preserving** distributed QA.
- Integrated **Meta LLaMA 3** with healthcare datasets for high-accuracy retrieval-based generation.
- Simulated federated learning environments using **Flower AI** for decentralized data processing.

Projects

School Management System (2000+ Users)

[Live](#) [✉](#) — [GitHub](#) [✉](#)

- Built a full-stack school management platform **serving 2000+ users** supporting admissions, results, and alumni networking.
- Implemented **Redis-based OTP authentication** and real-time chat using **Socket.IO**, reducing **40%** login latency.
- Integrated an **AI chatbot** using **LangChain + Retrieval-Augmented Generation (RAG)** for automated student and alumni support.

Governance Layer for Financial Agents (AgentOps)

[GitLab](#) [✉](#)

- Built a high-performance governance API gateway for autonomous AI agents with **sub-millisecond kill-switch** controls and real-time telemetry.
- Implemented granular permission scoping via **Open Policy Agent (OPA)** and dynamic spend caps using **Redis** in-memory evaluation.
- Engineered an autonomous AIOps remediation engine utilizing **LangGraph** and **Qdrant** vector memory.

Achievements

CodeChef: 4 star [codecrack2005](#) [✉](#)

AtCoder: 805 [sandipto729](#) [✉](#)

GeeksforGeeks: 700+ solved [sandipict3](#) [✉](#)

LeetCode: 1980+ Rating (Knight) [sandipto729](#) [✉](#)

Flipkart GRiD 7.0: National Semifinalist [Link](#) [✉](#)

JEE Adv/Main: AIR 6514 / 8012 (99.31%ile)

Amazon ML Summer School: [Certificate](#) [✉](#)

CodeFest '25: Top 900 / 20k+ [Link](#) [✉](#)

Position of Responsibility

Innovation and Incubation Cell (IIC), NIT Durgapur

Website Coordinator

- Developed and maintained the official [iic.nitdgp.in](#) [✉](#) platform, improving access to startup resources for students.
- Built and maintained the platform using **React, Node.js, Express, MongoDB, Redux, JWT authentication, and Azure Blob Storage**.
- Coordinated major innovation events including **Internal Smart India Hackathon, MSME Hackathon, Startup Summit, and AI2 Summit**.
- Organized entrepreneurship talks and guided students in **startup ideation and early-stage prototyping**.